A Spectrum Extension Approach for Radiometric Calibration of the Advanced Hyperspectral Imager Aboard the Gaofen-5 Satellite

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***Abstract*— The advanced hyperspectral imager (AHSI) is one of the sensors aboard the Chinese Gaofen-5 (GF-5) satellite, possessing characteristics of high spatial and spectral resolution, as well as width swath. To better understand the radiometric performance of GF-5/AHIS after its launch, this article presents an on-orbit radiometric calibration approach for AHSI visible and near-infrared (VNIR) and shortwave infrared (SWIR) sensors from field automatic observations with a field spectrometer in the absence of SWIR measurements. A spectrum extension method was proposed to extend the retrieved surface hyperspectral reflectance in the VNIR spectral ranges to SWIR by incorporating the historical hyperspectral reflectance library. The radiometric calibration coefficients of GF-5/AHSI were calculated by linear fitting of the observed digital number (DN) values with GF-5/AHSI and predicted at-sensor radiances with MODTRAN 5 based on extended hyperspectral surface reflectance. Comparisons with onboard calibration results were also performed, and the averaged relative differences were within 5% with 1δ standard deviations less than 10% for most bands, except for those in the atmospheric absorption and low signal-to-noise ratio bands. The comparison results indicate that the on-site radiometric calibration results are consistent with the onboard results, and the operational on-orbit radiometric calibration approach is reliable in the case that there are no measurements in the SWIR spectra range. The on-orbit radiometric performance of GF-5/AHSI rapidly degraded during the first several months after its launch and then tended to be relatively stable.**

***Index Terms*— Advanced hyperspectral imager (AHSI), calibration site, Gaofen-5 (GF-5) satellite, radiometric calibration, spectrum extension.**

I. Introduction

THERE have been an increasing number of satellites equipped with hyperspectral sensors that have been launched in recent years, due to the development of hyperspectral remote sensing imaging technology. Gaofen-5 (GF-5) is a hyperspectral satellite recently launched in China. One of the six payloads on the GF-5 satellite is an advanced hyperspectral imager (AHSI), which is composed of visible and near-infrared (VINR) sensors and shortwave infrared (SWIR) sensors. It has a total of 330 spectral bands ranging from 400 to 2500 nm, with spectral resolutions of 5 and 10 nm in the VNIR and SWIR, respectively [1]. The spatial resolution of GF-5/AHSI is 30 m, and the swath width of GF-5/AHSI is 60 km, which is wider than the swaths of the Hyperion [2] and compact high-resolution imaging spectrometer (CHRIS) [3] spaceborne hyperspectral sensors launched in the past few decades. Consequently, hyperspectral remote sensing images with rich spectral information were collected by GF-5/AHSI sensors. The availability of hyperspectral images can be widely used in land-use and land-cover change detection, agriculture and forest management, biochemical and biophysical property extraction of the Earth's surface, and so on [4]–[8]. For this, accurate and traceable radiometric calibration is required to achieve radiometric consistency of the hyperspectral images collected with past, current and future hyperspectral sensors.

Space-borne hyperspectral sensors were commonly spectrally and radiometrically calibrated in the laboratory before launch. However, the spectral and radiometric performance might change in space environments over time. Postlaunch radiometric calibration is very important to monitor radiometric changes of on-orbit hyperspectral sensors, and it is equally important to be able to place them on an absolute radiometric scale with SI traceability [9]. Vicarious radiometric calibration using *in situ* measurements over a ground reference calibration site or pseudoinvariant target or at the time of satellite overpass has been proven to be a good way to calibrate and evaluate the radiometric performance of spaceborne and airborne optical sensors [10]–[12]. It can also permit a traceable understanding of the radiometric stability of onboard sensors. Traditionally used vicarious radiometric calibration approaches need to collect the atmospheric and surface conditions of the pseudoinvariant calibration site at the time of satellite overpass of the calibration site [13]. However, the field campaigns are limited to several drawbacks, such as personnel availability, budget, and travel time to the calibration site. Furthermore, *in situ* measurements cannot be collected successfully due to bad weather and equipment failures, which means successful travel to the calibration site does not ensure successful *in situ* measurement collection [14]. It may also reduce the calibration frequency and cannot efficiently monitor the long-term radiometric performance changes of onboard sensors when using the traditional personal vicarious radiometric calibration method.

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Studies on automatic vicarious radiometric calibration based on collected *in situ* measurements in the absence of personnel have been carried out to overcome the drawbacks of the traditional personnel vicarious radiometric calibration approach [15]. The Remote Sensing Group (RSG) at the University of Arizona has made many efforts to develop the Radiometric Calibration Test Site (RadCaTS) at Railroad Valley, which has been successfully applied to the radiometric calibration of onboard sensors, such as Landsat-7/ETM+ [14], Landsat-8/OLI [16], Hyperion [16], moderate resolution imaging spectroradiometer (MODIS), Visible Infrared Imaging Radiometer Suite (VIIRS) onboard the Suomi National Polar-Orbiting Partnership spacecraft (SNPP/VIIRS), and Sentinel-2/MSI [17]. The Centre National D'Etudes Spatiales (CNES) and the European Space Agency (ESA) also developed automatic ground-based stations with a modified CE-318 sun photometer at La Crau and Gobabeb calibration sites, respectively [18], [19]. They were also successfully applied to SPOT and Sentinel-2/MSI radiometric calibration [18], [20]. The Baotou site in China was also deployed with an automated observation system using spectrometers with spectra ranges from 380 to 1080 nm and 2-nm sampling intervals [21]. It has also been used to radiometrically calibrate Landsat-8/OLI and Sentinel-2/MSI [22], [23]. Nevertheless, it cannot be used to calibrate sensors such as GF-5/AHSI in the spectral range of 1000–2500 nm due to the limited spectral range of spectrometers at the Baotou site.

In view of this, this study aims to propose an operational on-orbit radiometric calibration approach for AHSI VNIR and SWIR sensors from field automatic observations with a field spectrometer in the absence of SWIR measurements. A spectrum extension method was proposed to extend the retrieved surface hyperspectral reflectance in the VNIR spectral ranges to SWIR by incorporating the historical hyperspectral reflectance library. The proposed approach was then applied to the on-orbit radiometric calibration of GF-5/AHSI at the Baotou site. This article is organized as follows. Section II presents the method used for radiometric calibration and Section III presents the calibration site and the data acquisition. Section IV shows the results obtained for in-flight calibration. A discussion of the limitations of the radiometric calibration approach is presented in Section V. Finally, brief conclusions of this work are drawn in Section VI.

II. Method

A. Reflectance-Based Method for Radiometric Calibration

At-sensor absolute radiometric calibration is generally defined as the radiometric calibration coefficient determination achieved by connecting the digital number (DN) of the remotely sensed images to the at-sensor physical radiance [24]. For each spectral band of GF-5/AHSI, a linear relationship between the DN and at-sensor radiance can be expressed as follows:

$$L_{\text{at-sensor}} = a \cdot \text{DN} + b \tag{1}$$

where $L_{\text{at-sensor}}$ is the at-sensor radiance of GF-5/AHSI; DN is the digital value of GF-5/AHSI; and a and b are the gain and bias of the GF-5/AHSI absolute radiometric calibration coefficients, respectively. Then, the at-sensor apparent reflectance $[\rho_a(\theta_s, \theta_v, \phi_{s-v})]$ of GF-5/AHSI can be expressed as follows:

$$\rho_a(\theta_s, \theta_v, \phi_{s-v}) = \frac{\pi L_{\text{at-sensor}}}{\mu_s E_s} = \frac{\pi (a \cdot \text{DN} + b)}{\mu_s E_s} \tag{2}$$

where θ_s is the sun zenith angle; θ_v is the GF-5/AHSI viewing zenith angle; ϕ_{s-v} is the relative azimuth angle between the sun principal panel and the panel of observation; E_s is the exoatmospheric solar irradiance; and μ_s is the cosine value of the sun zenith angle.

Several vicarious calibration approaches, including reflectance-based, radiance-based, and irradiance-based approaches, have been proposed during the past few decades [25]–[27]. Among these approaches, the reflectance-based approach is the most commonly used vicarious radiometric calibration approach due to its relative calibration accuracy and operation. Thus, the reflectance-based vicarious calibration method is adopted in this study. The reflectance-based method predicts the apparent reflectance at the sensor from the measured surface reflectance using a radiative transfer model by configuring other necessary input parameters. In this way, the at-sensor apparent reflectance of GF-5/AHSI can be expressed with the radiative transfer equation as follows [28]:

$$\rho_a(\theta_s, \theta_v, \phi_{s-v}) = \rho_0(\theta_s, \theta_v, \phi_{s-v}) + \frac{\rho_t}{1 - \rho_t S} T(\theta_s) T(\theta_v) \tag{3}$$

where $\rho_0(\theta_s, \theta_v, \phi_{s-v})$ is the atmospheric reflectance resulting from the photons never interacting with the surface; ρ_t is the surface reflectance; $T(\theta_s)$ is the atmospheric transmittance between the sun and surface resulting from the effects of aerosol scattering and molecular absorption; $T(\theta_v)$ is the atmospheric transmittance between the surface and spaceborne sensor resulting from the effects of aerosol scattering and molecular absorption; and S is the atmospheric albedo.

Reflectance-based vicarious radiometric calibration is carried out by relating the at-sensor apparent reflectance of GF-5/AHSI according to (2) and (3). The atmospheric parameters (E_s, $\rho_0(\theta_s, \theta_v, \phi_{s-v})$, $T(\theta_s)$, $T(\theta_v)$, and S) in (2) and (3) needed to perform the reflectance-based vicarious radiometric calibration are predicted with the MODTRAN 5 radiative transfer code [29]. In this way, the gain and bias of the GF-5/AHSI absolute radiometric calibration coefficients can be obtained from all known atmospheric parameters, the surface reflectance of the calibration target, spectral response, and DN of the GF-5/AHSI images.

B. Spectrum Extension Method for Hyperspectral Surface Reflectance Determination

One of the most important measurements in the reflectance-based method is the surface reflectance measurement, which is usually performed in person by a well-calibrated spectrometer when the satellite passes the calibration site. An alternative way is to automatically obtain the surface reflectance with an unmanned observation system in the field. In this study, the hyperspectral surface reflectance (denoted ρ_{vnir}) of the desert target in the VNIR spectral ranges is retrieved first from

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the spectrometer measurements as follows:

$$\rho_{\text{vnir}} = \frac{\pi L_t}{\mu_s E_S \left[e^{-\tau/\mu_s} + t_d(\theta_s)\right]} \tag{4}$$

where L_t is the obtained radiance of the desert target from the spectrometer of the automatic observation system and $e^{-\tau/\mu_s}$ and $t_d(\theta_s)$ are the direct and diffuse atmospheric transmittances from the sun to the surface, respectively.

An operational spectrum extension method is proposed to calibrate GF-5/AHSI VNIR and SWIR sensors from automatic field observations with a field spectrometer in the absence of SWIR measurements. The retrieved hyperspectral surface reflectance in the VNIR spectral ranges is extended to SWIR by incorporating the historical hyperspectral reflectance library of the desert target. The historical hyperspectral reflectance library was built based on the hyperspectral reflectance collected during past on-site radiometric calibration campaigns under clear sky conditions. All hyperspectral reflectance were measured with an SVC HR-1024 spectroradiometer that was radiometrically calibrated at the National Institute of Metrology (NIM) of China. Therefore, the main uncertainty of the spectral library is due to the radiometric calibration uncertainty of the SVC HR-1024 spectroradiometer at the NIM, which was estimated to be less than 2% ($K = 1$) according to the reported certification. Then, for each hyperspectral surface reflectance (denoted j) in the spectral library, we calculate the mean absolute error (MAE) with the retrieved hyperspectral surface reflectance in the VNIR spectral range as follows:

$$\text{MAE}_j = \frac{\sum_{i=1}^{n} |\rho_{\text{vnir},i} - \rho_{\text{lib},i,j}|}{n} \tag{5}$$

where $\rho_{\text{lib},i,j}$ is the hyperspectral surface reflectance at band i. Note that the MAE was calculated only in the spectral range of 500–750 nm for the following two reasons. One reason is the low signal-noise ratio of the collected data in the spectral ranges of 380–500 and 750–1080 nm. The other reason is that the collected data in the spectral range of 500–750 nm are less sensitive to the environmental temperature. Then, we select the historical spectral library, with hyperspectral surface reflectance acting as a reference, when the calculated MAE is the minimum. The minimum MAE is specified as the spectrum extension factor from the VNIR spectral ranges to the SWIR spectral range. In this way, the surface hyperspectral surface reflectance in the visible, near infrared, and SWIR spectral ranges (denoted $\rho_{\text{vnir-swir}}$) can then be determined as follows:

$$\rho_{\text{vnir-swir}} = \rho_{\text{lib},j} + \text{MAE}_j. \tag{6}$$

III. Calibration Site and Dataset

A. Calibration Site

It is very important to select a calibration site for vicarious radiometric calibration. An ideal calibration site should usually be qualified with several characteristics summarized by Thome [27]. For example, a high reflectance greater than 30% ensures that the sensor-observed radiance is mainly contributed to the surface-reflected solar irradiance. In addition, high spatial uniformity, spectral uniformity, and stability should also be considered when selecting the calibration site. In this study, a large-area natural desert target at the Baotou calibration site was selected as our vicarious radiometric calibration site. The geographical coordinates of the Baotou calibration site are 40.88°N, 109.53°E, and the site is approximately 50 km from Baotou city of Inner Mongolia in northern China, with an average elevation of approximately 1270 m above sea level. The size of the desert target is large enough to ensure that no less than ten pixels of GF-5/AHSI have a 30-m spatial resolution. The surface reflectance of the desert target is generally larger than 0.3.

TABLE I
Acquisition Information of GF-5/AHSI Images and Atmospheric Parameters at the Baotou Calibration Site

	2018/8/8	2018/11/25	2019/10/11	2020/3/26
UTC Time	5:46:24	5:42:54	5:35:59	5:35:50
SZA (°)	27.74	63.83	50.1	39.78
VZA (°)	0.55	0.22	0.08	0.08
SAA (°)	353.56	199.54	202.34	198.97
VAA (°)	212.02	177.16	158.51	158.7
AOT @ 550 nm	0.072	0.1671	0.0932	0.1752
WVC (g/cm²)	1.8616	0.3099	0.4988	0.2739

B. GF-5/AHSI Images

The GF-5 satellite is designed to fly in a sun-synchronous orbit at an altitude of 705 km, and the revisit time with a nadir viewing angle of the GF-5 satellite is 51 days. Four GF-5/AHSI hyperspectral images with a near nadir viewing angle under clear sky conditions were collected successfully over the Baotou calibration site since it was launched. The acquisition information, including the overpass date, UTC time, solar and viewing zenith angles, and solar and viewing azimuth angles, is listed in Table I. Fig. 1(a) shows a subset RGB composite hyperspectral 3-D cube of the Baotou site extracted from GF-5/AHSI hyperspectral images collected on October 11, 2019.

C. Ground Measurements

The surface characteristics and synchronous atmospheric parameters are the most important parameters measured in vicarious radiometric calibration. At the Baotou site, two automated observation systems with spectrometers, as shown in Fig. 1(b), are used to measure the reflected radiance of the desert target every 2 minutes. The spectrometers cover a spectral range of 380–1080, with a spectral sampling interval of 2 nm. An automated CIMEL sun photometer is used to measure the atmospheric parameters of the Baotou calibration site, which is one of the operational AErosol RObotic NETwork (AERONET) sites, called "AOE_Baotou." The direct solar and sky irradiance datasets collected throughout the day by the sun photometer are processed once they are uploaded to the AERONET data process center. The processed atmospheric parameters, including aerosol optical thickness and columnar water, are available on the AERONENT website. These collected surface and atmospheric parameters are used in the radiative transfer calculations. In addition, the *in situ*-measured surface reflectances of the desert target were also

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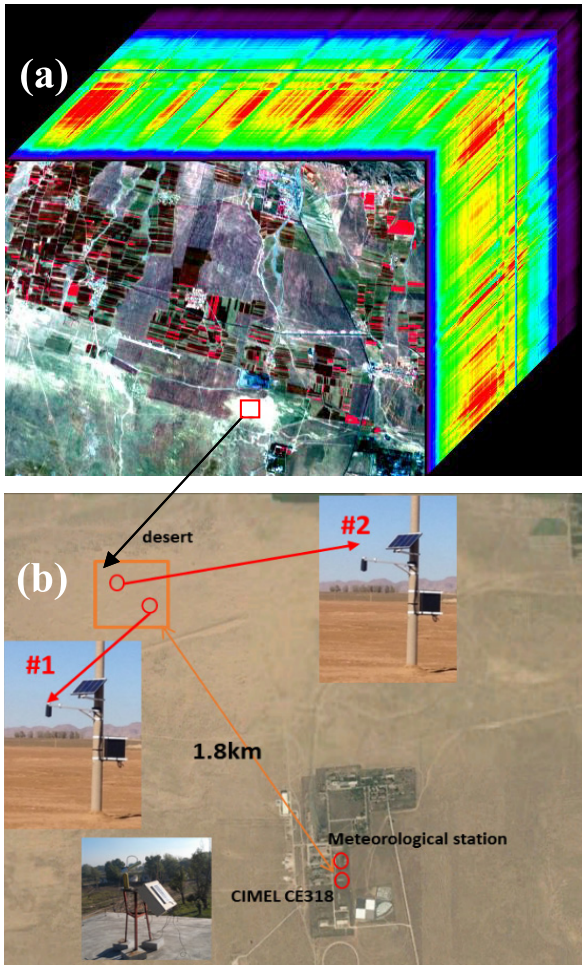


Fig. 1. (a) Subset RGB composite hyperspectral 3-D cube of the Baotou site extracted from GF-5/AHSI hyperspectral images collected on October 11, 2019. (The red box indicates the position of the desert target.) (b) Ground automated observation system deployed at the Baotou site.

collected with an SVC HR-1024 spectroradiometer when the GF-5 satellite passed over the Baotou calibration site on October 11, 2019.

IV. Results

A. Surface Reflectance Spectra Results

According to the method described in Section II, the surface reflectance spectra of the desert in the VNIR and SWIR spectral range at the time of the four GF-5 satellite overpass dates were determined and are shown in Fig. 2. The standard deviation of the four surface reflectance spectra was calculated and is shown with the dashed line with purple color in Fig. 2. The standard deviation is generally less than 0.015. The averaged standard deviation in the spectral range of 400–2500 nm is 0.0097. The results suggest that the surface reflectance characteristic of the desert target is relatively stable and is suitable for space-borne sensor radiometric calibration.

B. Radiometric Calibration Results

Automated radiometric calibration was successfully implemented four times for GF-5/AHSI at the instrumented Baotou site over the 2018–2020 period. The MODTRAN 5 radiative

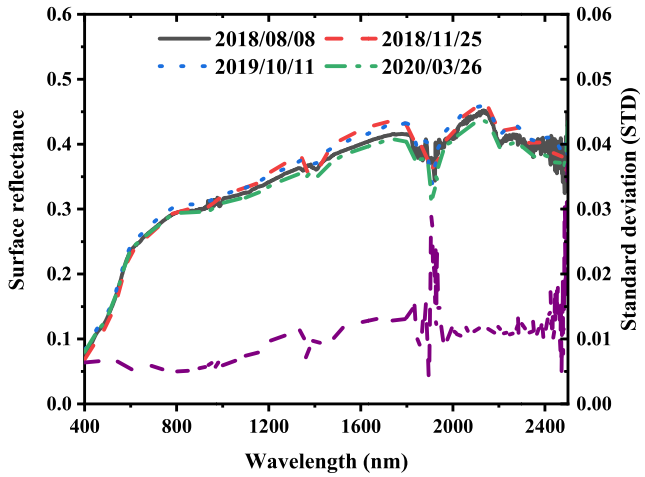


Fig. 2. Hyperspectral surface reflectance spectra of the desert target determined when the GF-5 satellite passed over the Baotou site on the four satellite overpass dates.

transfer code was configured with estimated surface reflectance spectra and corresponding atmospheric parameters used to predict at-sensor radiances. Then, the radiometric gains of VNIR and SWIR sensors were calculated by combining the predicted radiances and extracted DN values from the GF-5/AHSI images. Fig. 3 shows the radiometric calibration results of GF-5/AHSI obtained with our proposed approach. The results for spectral ranges of 1350–1420 and 1800–1940 nm are missing because the DN values extracted from GF-5/AHI images are unavailable. Although ground-measured atmospheric parameters were used to predict at-sensor radiances, slight fluctuations were observed at the atmospheric absorption bands, such as the oxygen bands at approximately 760 nm, the carbon dioxide bands near 2010 and 2060 nm, and, especially, the water vapor bands centered near 940 and 1140 nm. The results responsible for this fluctuation in radiometric calibration might be caused by the spectral shift of GF-5/AHSI.

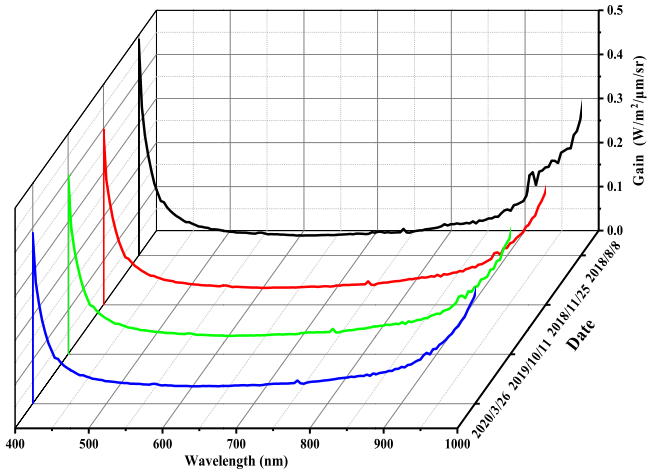
C. Comparison Results With Onboard Calibration

The comparison results are presented as the relative difference (denoted RE) between the at-sensor radiances (denoted L) of the desert target at the Baotou site calculated with onboard and on-site calibration coefficients, and the relative difference is calculated as follows:

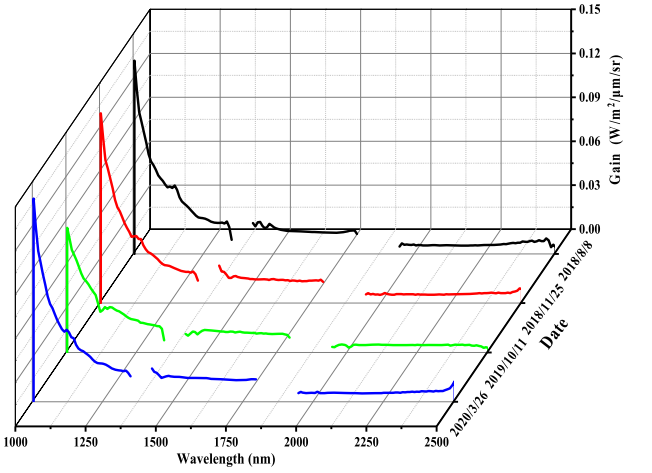
$$\mathrm{RE} = 100\% \times \frac{L_{\text{on-site}} - L_{\text{on-board}}}{L_{\text{on-board}}}. \tag{7}$$

Fig. 4 shows the relative difference between on-site and onboard calibration results on the four calibration dates. Generally, the comparison results show a smaller change in the VNIR region than in the SWIR region. Large relative differences occur near 760 and 940 nm in the VNIR spectral range and near 1140, 2010, and 2060 nm in the SWIR spectral range. These are mainly affected by atmospheric absorption by oxygen, water vapor carbon dioxide, and so on. Moreover, the low reflected energy at atmospheric absorption bands results in a larger calculated relative difference. Fig. 4 also shows that the trends of the relative differences between the on-site and

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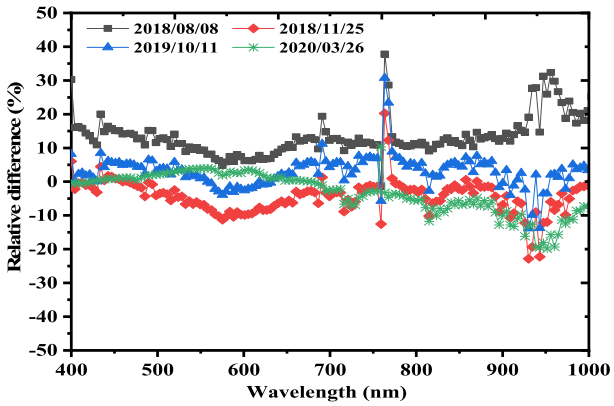


(a)

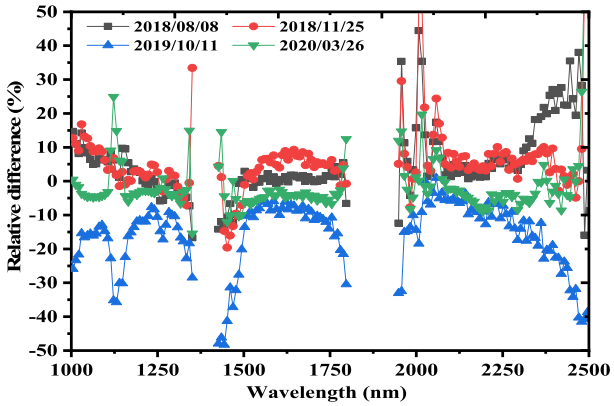


(b)

Fig. 3. On-site automated radiometric calibration coefficients of GF-5/AHSI. (a) VNIR sensor. (b) SWIR sensor.



(a)



(b)

Fig. 4. Comparison results of relative differences between on-site and onboard radiometric calibration on the four calibration dates. (a) VNIR sensor. (b) SWIR sensor.

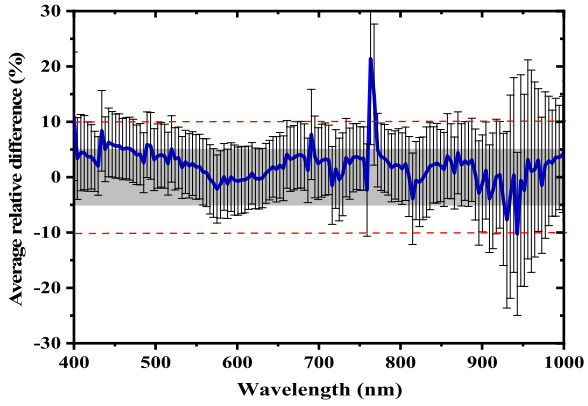
onboard calibration results are very similar in the VNIR region on August 8, 2018, November 25, 2018, and October 11, 2019. However, the trend of the relative differences on March 26, 2020 is different from that of the previous three dates. This can be explained by the updated onboard calibration results on March 26, 2020. However, there are no obvious similar trends of relative differences for the SWIR sensor, especially for the comparison results on October 11, 2019. This might be caused by the onboard radiometric performance change for the SWIR sensor after almost a year and a half of flight in the space environment. The relative differences are closer to zero for most of the bands than the updated onboard calibration results on March 26, 2020. This demonstrates that frequency and lifetime radiometric calibration of the onboard sensor is very important for the further quantitation of the collected GF-5/AHI images.

Fig. 5 shows the relative differences between the desert TOA radiances obtained by on-site and onboard calibration by averaging the comparison results from the four calibrations. Excluding the high variability of the spectral regions affected by atmospheric absorption, the averaged relative differences between the radiances of the four calibration dates are within 5% for both the VNIR and SWIR sensors. The error bars shown in Fig. 5 are the 1σ standard deviation of the relative differences between the on-site and onboard desert TOA radiance values on the four calibration dates. The average relative differences with 1δ standard deviations are less than 10% for most bands except for those in atmospheric absorption. Nevertheless, the average relative differences are slightly higher in the SWIR range than in the VNIR range, which might be due to the strong atmospheric absorption of gas molecules as well as the low signal-to-noise ratio. The uncertainty of estimated surface reflectance in the SWIR spectra range with the method introduced in this study could also result in the uncertainty of radiometric calibration. The on-site radiometric calibration results are consistent with the onboard radiometric calibration results, which also demonstrates the reliability of the on-site radiometric calibration results obtained using the operational method introduced in this study.

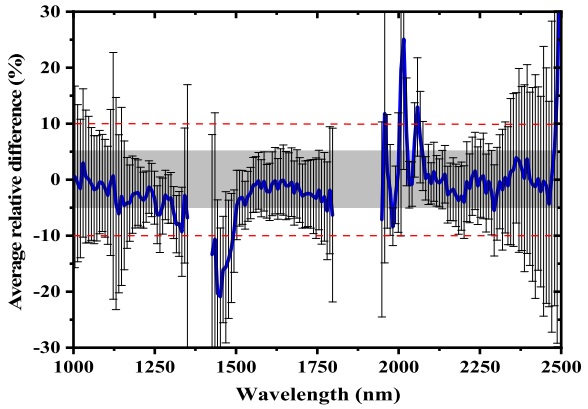
D. ON-Orbit Radiometric Performance Degradation Analysis

Further quantitative applications, such as land cover change detection, usually rely on long-term hyperspectral images

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(a)



(b)

Fig. 5. Average relative differences between on-site and onboard radiometric calibrations for the four calibration days and the error bar showing the 1δ standard deviation of the relative differences for the four days. (a) VNIR sensor. (b) SWIR sensor.

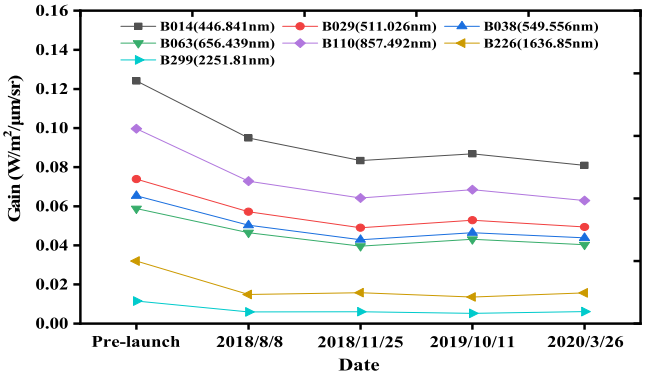


Fig. 6. Degradation of GF-5/AHSI sensor radiometric performance, with gains at bands corresponding to the center wavelength of Landsat-8/OLI.

collected with GF5/AHI. Therefore, it is very important to understand whether any degradation has occurred on board GF5/AHI sensors. Fig. 6 shows the degradation of the GF5/AHSI SWIR sensor radiometric calibration coefficient, with gains at bands corresponding to the center wavelength of Landsat-8/OLI. Rapid degradation occurred for the first on-site radiometric calibration performed on August 8, 2018 compared to the prelaunch calibration results obtained in the laboratory. The radiometric calibration coefficients with gains on August 8, 2018 changed from the prelaunch calibration results in the laboratory by 23.4% and 50.9% for the VNIR and SWIR sensors, respectively. This is mainly due to the environmental change in space after launch in the test operation phase of GF5/AHSI. After that, the radiometric calibration coefficient decreased slightly on November 25, 2018, with degradations of 13.6% and 39.4% for the VNIR and SWIR sensors, respectively. Then, the radiometric calibration coefficients tended to be stable in the following two on-site calibrations on October 11, 2019 and March 26, 2020. The average changes in these two calibrations are 7.18% and 6.73% for the VNIR sensor compared with the radiometric calibration results obtained on November 25, 2018. The average changes for the SWIR sensor are 14.1% and 16.4%. The radiometric performance degradation analysis results demonstrate that the radiometric performance tends to be stable after November 25, 2018, after several months of GF-5/AHSI in-orbit tests in space.

V. Discussion

A. Spatial Nonuniformity of the Calibration Site

Unlike traditional on-site personnel *in situ* measurements, spatial nonuniformity effects could be greatly reduced due to high spatial sampling [16]. Spatial sampling in automatic ground measurements is limited to the number of deployed spectrometers. In this study, only two spectrometers were used to measure the reflected radiance over the desert target at the Baotou calibration site. This means that the automatic ground measurements were not truly representative of the whole surface reflectance over the whole large desert area. To understand the representativeness of the two spectrometer measurements for the whole site, the spatial uniformity of the desert target at the Baotou calibration site was analyzed based on the GF-5/AHSI hyperspectral images collected on the four calibration dates. The collected hyperspectral images were downscaled from a spatial resolution of 30 m to 0.13 m, which is the viewing area of the spectrometers on the ground. The spatial nonuniformity of the desert target is evaluated with the coefficient of variation (denoted CV), which is defined as follows:

$$\mathrm{CV}(\lambda) = 100\% \times \frac{\sigma(\lambda)}{\mu(\lambda)} \tag{8}$$

where $\sigma(\lambda)$ and $\mu(\lambda)$ are the standard deviation and mean of the values over the whole desert target, respectively. Fig. 7 shows the spatial nonuniformity, with the calculated CV based on downscaled hyperspectral images. The results indicate that the averaged uncertainties in the VNIR spectral range resulting from the spatial nonuniformity of the desert target are 2.11%, 1.38%, 1.18%, and 0.88% on the four calibration dates if only one spectrometer is used. According to a reported study [30], the uncertainties are averaged by using two spectrometers for measurements and yield uncertainties of 1.49%, 0.97%, 0.83%, and 0.63% on the four calibration dates. The spatial nonuniformity analysis results were encouraging because they demonstrated that the good spatial uniformity of the desert

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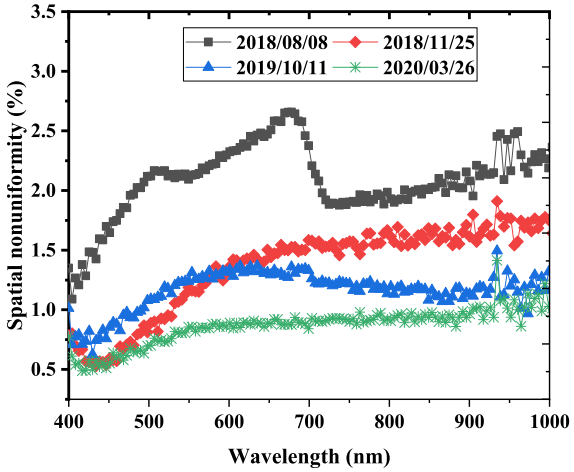


Fig. 7. Evaluation of the spatial nonuniformity of the desert target with coefficients of variation using spatial downscaled hyperspectral images.

target at the Baotou site and the two spectrometers were generally sufficient to sample the desert target. In addition, the spatial nonuniformity on August 8, 2018 was slightly larger than that on the other three days, which was most likely due to sparse grass changes in and crops around the desert site during the vegetation growth period.

B. Implication of Field Automatic Observations

The implication of automatic field observations is mainly associated with the temperature sensitivity of the spectrometers. The spectrometers deployed over the desert target at the Baotou calibration site were calibrated every year at the NIM of China. The calibration coefficients of the spectrometers were measured in the NIM laboratory at an indoor temperature of 25 °C. However, the calibration coefficients of the spectrometers could change due to environmental temperature changes in the field conditions at the Baotou calibration site. In view of this, an empirical model, as expressed in (9), was established to characterize the changes in the calibration coefficients [denoted $f(\lambda, T)$] of the spectrometers as a function of wavelength and temperature [30]. The unknown parameters (denoted a_i and b_i) in (9) were determined by fitting the measured datasets in the laboratory, as shown in Fig. 8(a), which was obtained by calibrating the spectrometers against a stable source while its temperature was varied from 10 °C to 40 °C, and the results were compared to the reference calibration results obtained at 25 °C

$$f(\lambda, T) = \left(a_0 + a_1\lambda + a_2\lambda^2\right)T + \left(b_0 + b_1\lambda + b_2\lambda^2\right). \quad (9)$$

Two components should be considered in the temperature sensitivity of the spectrometers. One component is the error of the empirical model, which means whether the empirical model could characterize the actual temperature sensitivity of the spectrometers. It was estimated from the residual between the results measured in the laboratory and predicted results with an empirical model. The other component is the error in the temperature measurement of the spectrometer, which

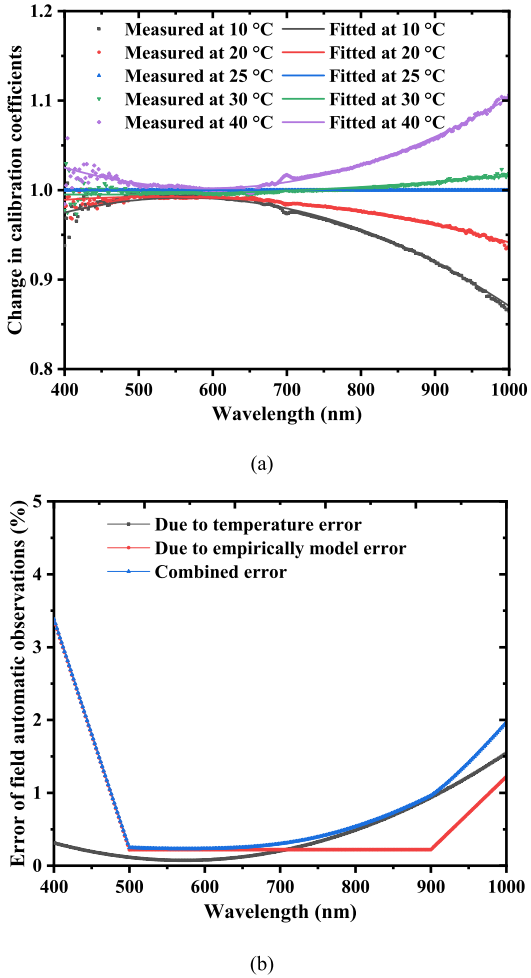


Fig. 8. (a) Changes in the calibration coefficients of the spectrometers as a function of wavelength based on data sets measured in the laboratory. (b) Combined error of automatic field observations due to temperature and empirical model errors.

propagated to the field observations. Fig. 8(b) shows the combined error of automatic field observations due to temperature measurement and empirical model errors. Note that the error declines rapidly with wavelengths from 400 to 500 nm, remains relatively stable from 500 to 750 nm, and increases smoothly in the spectral range of 750–1000 nm. The error of automatic field observations due to the temperature sensitivity of spectrometers in the spectral range of 500–750 nm is less than 0.4%, which suggests that the automatic field observations of spectrometers are less sensitive to temperature changes in these spectral ranges. This can explain why only the surface reflectance retrieved in the spectral range of 500–750 nm was used to determine the surface hyperspectral reflectance from the historical spectral library of reflectance, as described in Section II-B.

C. Influence of Surface Reflectance Determination

In this study, the surface hyperspectral reflectance of the desert in the VNIR spectral ranges was calculated first from automatic field observations with a spectrometer and

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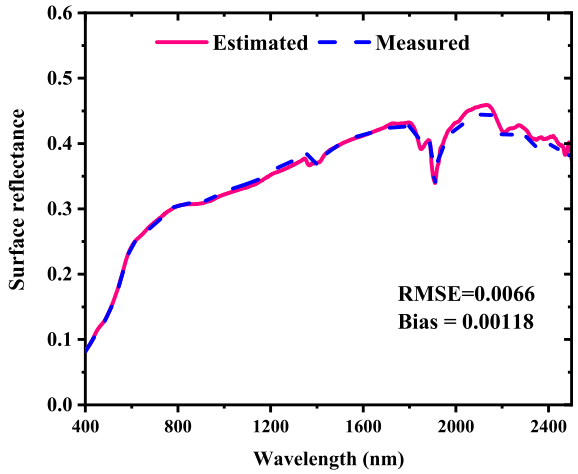


Fig. 9. Validation of estimated and measured surface reflectances of the desert target when the GF-5 satellite passed over the Baotou site on October 11, 2019.

diffuse sky irradiance predicted with MODTRAN 5. Therefore, the influence of surface reflectance determination resulting from the uncertainty of predicted diffuse sky irradiance in addition to that of automatic field observations should be considered. The overall uncertainty of predicted diffuse sky irradiance was estimated to be less than 3.1%, mainly related to the uncertainty of the MODTRAN model, water vapor content, aerosol optical thickness, aerosol types, and solar irradiance [31]. The accuracy of the diffuse sky irradiance prediction and effect on surface reflectance retrieval were also analyzed by comparing with ground measurements, and the error was estimated to be greater than 10% at 400 nm [32]. Furthermore, the influence of predicted diffuse sky irradiance tends to increase with decreasing wavelength due to the aerosol scattering effect.

There is another limitation to hyperspectral surface reflectance determination in the spectral range of GF5/AHSI sensors. The hyperspectral surface reflectance used to predict the GF5/AHSI at-sensor radiance is extended using the hyperspectral reflectance retrieved in the VNIR spectral range from the spectrometer. This spectrum extension does not consider any changes in the spectral shape of the desert target at the Baotou calibration site, which means that the proposed spectrum extension method assumes that the spectral shape of the desert target at the Baotou calibration site is relatively stable. Moreover, there is no reference in the shortwave spectral range. Validation of the estimated surface reflectance spectrum was carried out by comparing to *in situ* measurements obtained on October 11, 2019, as shown in Fig. 9. There is a high degree of consistency between the spectral value and shape of the estimated and measured surface reflectance spectra. The calculated root means square error (RMSE) and bias are 0.0066 and 0.0018, respectively. However, the differences beyond the wavelength of 2200 nm are slightly larger than those below 2200 nm, which may result from the low signal-noise ratio of the spectroradiometer in the spectral range beyond 2200 nm. The limitation of the spectrum extension method could also result in the error in the hyperspectral surface reflectance determined in the SWIR spectral range.

VI. Conclusion

This article presents an on-orbit radiometric calibration approach for AHSI VNIR and SWIR sensors from automatic field observations with field spectrometers in the absence of SWIR measurements. The hyperspectral surface reflectance was retrieved first from automatic field observations and modeled diffuse sky irradiance. Then, a spectrum extension method is proposed to extend the retrieved surface hyperspectral reflectance in the NNIR spectral ranges to SWIR by incorporating the historical hyperspectral reflectance library of the desert target. The proposed approach was applied to the radiometric calibration of the four GF-5/ASHI images collected while passing the Baotou calibration site since the satellite was launched on May 9, 2018. The radiometric coefficients of GF-5/AHSI were calculated by linear fitting the observed DN values with GF-5/AHSI and predicted at-sensor radiances with MODTRAN 5 based on the extended hyperspectral surface reflectance. Comparing results with onboard calibrations shows that the averaged relative differences on the four calibration dates are within 5%, with 1δ standard deviations less than 10% for most bands except for those in the atmospheric absorption and low signal-to-noise ratio bands. The on-orbit radiometric performance of GF-5/ rapidly degraded during the first several months after its launch and then trended to be relatively stable.

Furthermore, limitations of the proposed on-site radiometric calibration approach were discussed in this article. First, the spatial nonuniformity was estimated to be 1.49%, 0.97%, 0.83%, and 0.63% with two automatic field spectrometers present over the desert target at the Baotou calibration site. Second, automatic field observations with spectrometers between 500 and 750 nm were found to be less temperature-sensitive, and the uncertainties were estimated to be less than 0.4%. Finally, the influence of surface reflectance determination due to predicted diffuse sky irradiance was discussed and estimated to be less than 3.1% according to the authors' previous studies. Moreover, the hyperspectral surface reflectance errors determined with the proposed spectrum extension method were also evaluated, and the RMSE and bias were 0.0066 and 0.0018, respectively. Note that the BRDF effect of the calibration site is not considered in this study due to the near nadir view zenith angle of GF-5/AHSI. However, BRDF cannot be neglected when GF-5/AHSI is not in the nadir view zenith angle. Future work will focus on evaluating the relationship between spectrometer numbers and their possible locations and spatial nonuniformity. Ground automatic measurements of the diffuse sky irradiance arriving on the surface will also be considered in future work to improve the accuracy of surface reflectance retrieval.

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